

Deep Learning for Solving Dynamic Models

Session 7 — Structural estimation via simulated method of moments

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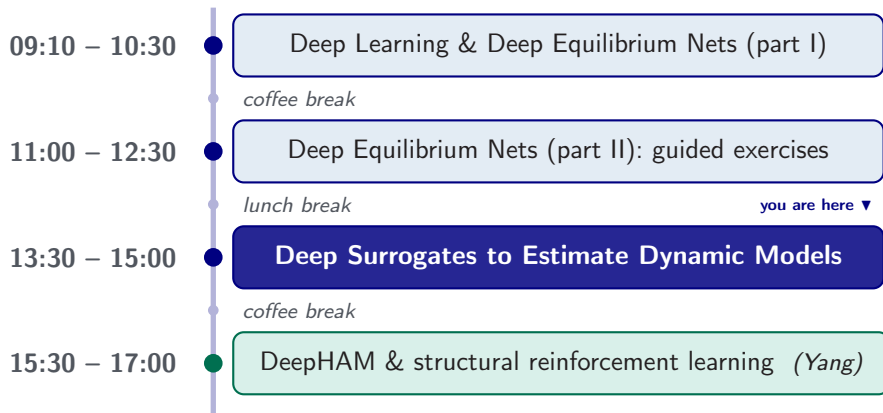
HEC, University of Lausanne

Based on Chen, Didisheim & Scheidegger (2026), *Journal of Financial Economics* 177, 104222

University of Padova

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Day 1 at a Glance



Materials: github.com/sischei/deep_learning_padova_0926 · Cloud: nuvolos.cloud

Roadmap of This Lecture (90 min)

I. What Is a Surrogate? (12 min)

- ▶ Look-up tables, pseudo-states, and what the speed-up buys

II. Estimate: structural estimation (20 min)

- ▶ SMM on a surrogate; Brock–Mirman; the identification ridge

III. Gaussian Processes (24 min)

- ▶ When data is expensive, a deep net is the wrong tool
- ▶ Kernels, credible bands, and active learning

IV. Quantify: uncertainty (10 min)

- ▶ Univariate effects, Sobol' indices, Shapley values for the SCC

V. Design: the optimal carbon tax (16 min)

- ▶ Policy coefficients as pseudo-states; 500 solves, not a million

VI. Synthesis and Hands-On (8 min)

- ▶ When to use what; the notebooks

Learning Objectives

By the end of this lecture you should be able to:

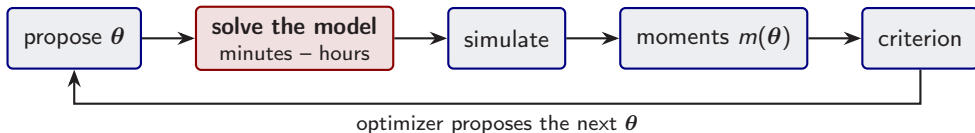
- ▶ Explain a surrogate as a **differentiable look-up table** and say precisely which classical bottleneck it removes
- ▶ Promote structural or policy parameters to **pseudo-states**, so that one solve answers a whole family of questions
- ▶ Estimate a structural parameter by simulated method of moments on a trained surrogate, and read an identification ridge off the criterion surface
- ▶ Turn the same surrogate into a global sensitivity analysis: univariate effects, Sobol' indices, Shapley values
- ▶ Fit a Gaussian process, read its credible band, and **decide between a GP and a deep network** from the cost of one model evaluation
- ▶ Place the next expensive evaluation where it buys the most, using Bayesian active learning

Part II

Estimate: Structural Estimation

The Estimation Loop, and Why It Is Expensive

The classical nested fixed point. For each candidate θ :



The surrogate move. Put θ *inside* the network at training time. The red box then runs **once**, and the loop becomes a forward pass.

Two consequences, not one

- ▶ The criterion becomes cheap enough to evaluate on a fine grid, so you can *look at* it rather than trust a local optimizer's report.
- ▶ The criterion becomes **differentiable in θ** , because the surrogate is.

SMM: The Moment Condition

Let the data give T observations $\{y_t\}$. Choose moment functions $g(y) \in \mathbb{R}^q$ such that, at the true parameter $\theta_0 \in \mathbb{R}^p$ with $q \geq p$,

$$\mathbb{E}[g(y_t) - m(\theta_0)] = 0, \quad m(\theta) = \mathbb{E}_\theta[g(y_t)].$$

The empirical and simulated counterparts are

$$\hat{g}_T = \frac{1}{T} \sum_{t=1}^T g(y_t), \quad \hat{m}_S(\theta) = \frac{1}{S} \sum_{s=1}^S g(y_s^{\text{sim}}(\theta)).$$

Identification. θ_0 is the *unique* zero of $\mathbb{E}[g(y_t)] - m(\theta)$ on the parameter space.

Just-identified: $q = p$. Over-identified: $q > p$, which buys a testable restriction. It does *not* automatically sharpen the criterion: a moment with a near-zero column in M adds noise and finite-sample bias, as the joint exercise will show.

In this lecture. y_t is log output, consumption, and capital; g collects three volatility and autocorrelation moments; $\theta = \varrho$ first, then $\theta = (\beta, \varrho)$.

SMM: Estimator and Weighting

$$\hat{\theta}_{\text{SMM}} = \arg \min_{\theta} [\hat{g}_T - \hat{m}_S(\theta)]^\top W [\hat{g}_T - \hat{m}_S(\theta)], \quad W \succ 0.$$

Common random numbers: the detail that decides success

Hold the shock draws $\{\varepsilon_s\}$ **fixed across θ** . Then $\hat{m}_S(\theta)$ is a smooth function of θ , with a clean minimum. Re-draw them and you are optimizing a noisy surface: the optimizer chases simulation noise and reports wherever it happened to land.

Weighting matrix.

- ▶ **Identity** $W = I$: the honest pedagogical default; consistent, inefficient.
- ▶ **Diagonal** $W = \text{diag}(1/\hat{\sigma}_g^2)$: rescales each moment by its sampling variance.
- ▶ **Optimal** $W^* = \Omega^{-1}$, the inverse asymptotic covariance of the moment gap, proportional to $\hat{\Sigma}_g^{-1}$ up to the simulation factor $1 + 1/S$. Minimum asymptotic variance. ($S = 10$ costs 5% in standard errors; $S = 1$ costs 41%.)

SMM: What the Asymptotics Tell You to Look At

Write $M = \partial m(\theta_0)/\partial \theta^\top$ for the moment Jacobian, and Ω for the asymptotic covariance of the **moment gap** $\sqrt{T}[\hat{g}_T - \hat{m}_S(\theta_0)]$. If the simulated shocks are drawn independently of the data, $\Omega = (1 + 1/S)\hat{\Sigma}_g$ with S panels of the data's length; as $S \rightarrow \infty$, $\Omega \rightarrow \hat{\Sigma}_g$. **The inflation is the price of simulating rather than solving.**

Under the usual regularity conditions, as $T \rightarrow \infty$,

$$\sqrt{T}(\hat{\theta}_{\text{SMM}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V(W)), \quad V(W) = (M^\top W M)^{-1} M^\top W \Omega W M (M^\top W M)^{-1},$$
which collapses to $V(W^*) = (M^\top \Omega^{-1} M)^{-1}$ at the efficient weighting.

Identification, read geometrically

V exists only if M has **full column rank**. A near-rank-deficient M is a **flat direction** of the criterion: an *identification ridge*. Standard errors explode, the point estimate is arbitrary along the ridge, and no amount of data at those moments fixes it.

Because the surrogate makes the criterion cheap, you can **plot the ridge** instead of inferring it from an ill-conditioned Hessian. We do exactly that at the end of this part.

The Surrogate Is Not Free

Everything on the previous slide assumed the moments came from *the model*. They came from a network. Write $\hat{m} = m + e$, where e is the surrogate's approximation error.

e does not shrink with the sample

More data makes $\hat{g}_T$ precise; it does nothing to e . This is **non-local misspecification**: the estimator converges not to θ_0 but to a **pseudo-true** θ^* minimizing the population criterion under $\hat{m}$, with Hall–Inoue asymptotics rather than Hansen's.

$$\sqrt{T}(\hat{\theta} - \theta^*) \xrightarrow{d} \mathcal{N}(0, \Sigma(\mu^*)), \quad \theta^* \rightarrow \theta_0 \text{ only as } \sup_{\theta} \|e(\cdot, \theta)\| \rightarrow 0.$$

The practical consequence, and the whole reason Part I ended with a validation step: report the surrogate's **maximum** error over the parameter box, not its mean. The maximum bounds the bias, and a surrogate with excellent *average* accuracy can still be badly wrong in the one direction you are estimating. That happens at the end of this part.

Chen, Didisheim & Scheidegger (2026), fn. 15, make the point about their own surrogate; the asymptotics are Hall & Inoue (2003).

The Model: Brock–Mirman with Stochastic TFP

The same model class as this morning, with two deliberate changes: δ drops from 1 to 0.10, which destroys the closed form, and ϱ is promoted from a fixed number to an input.

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t], \quad K_{t+1} + C_t = Y_t + (1 - \delta)K_t,$$

$$Y_t = z_t K_t^\alpha, \quad \log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1).$$

Calibration

α	δ	σ	β (scalar exercise)
0.36	0.10	0.04	0.96

Note the δ . The textbook closed form $s^* = \alpha\beta$ needs $\delta = 1$. Here $\delta = 0.10$, so there is *no* analytic policy and the Euler equation must be solved numerically, which is precisely the expensive inner solve the surrogate replaces. Choosing $\delta = 1$ would have made the exercise vacuous.

Why ϱ , and Not β ?

- ▶ β is famously **poorly identified from business-cycle moments**: volatilities and autocorrelations barely move with it. It does shift the *level* of the capital stock, which is the channel we exploit at the end of this part.
- ▶ In practice β is *calibrated* to a target steady-state interest rate. Under log utility with general δ , the steady-state Euler relation is $1/\beta = 1 + \alpha(k^*)^{\alpha-1} - \delta$.
- ▶ ϱ has a **clean, direct mapping** into autocorrelation moments: $\text{Corr}(\log Y_t, \log Y_{t-1})$ and $\text{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$ both move monotonically with persistence.
- ▶ So ϱ is the textbook **sharply identified** scalar, what you want when learning the workflow. Three moments for one parameter is over-identified, which buys a testable restriction; it does *not* automatically sharpen anything.

Notebook 03_03 does the scalar case. Notebook 03_04 then frees β as well, and the pathology shows up immediately.

Step 1: Train the Surrogate with ϱ as a Pseudo-State

Goal

Learn $\phi : (z_t, K_t, \varrho) \mapsto s_t \in (0, 1)$, with $K_{t+1} = (1 - \delta)K_t + s_t Y_t$.

1. A three-input MLP with a *sigmoid* output head, so $s_t \in (0, 1)$ holds by construction: the hard-constraint trick from this morning.
2. Train on the Euler residual with Gauss–Hermite quadrature. ϱ enters both the policy and the transition. Under log utility the notebooks minimize the scale-free *relative* residual, which shares its zero set with the raw one but conditions far better in FP32:

$$\ell = \frac{1}{N} \sum_i \left(\left[\beta C_t^{(i)} \sum_j w_j \frac{1 - \delta + \alpha z_{t+1,j}^{(i)} K_{t+1}^{(i)\alpha-1}}{C_{t+1,j}^{(i)}} \right]^{-1} - 1 \right)^2$$

3. Two phases: uniform sampling over the (z, K, ϱ) box, then simulation-based refinement.
4. Validate by plotting $s(z=1, K, \varrho)$ across ϱ , and ask first how much the policy *should* move with ϱ . Here the honest answer is “not much”, which is worth knowing *before* you diagnose lazy learning.

Step 2: Simulate, and Choose Moments That Identify

Goal

Generate synthetic data at $\varrho_{\text{true}} = 0.90$; three identifying moments, plus a counterexample.

Moment	What it captures
$m_1 = \text{Std}(\Delta \log C_t)$	growth-rate volatility
$m_2 = \text{Corr}(\Delta \log C_t, \Delta \log C_{t-1})$	persistence in consumption growth
$m_3 = \text{Corr}(\log Y_t, \log Y_{t-1})$	persistence in output, the most direct
$m_0 = \text{mean}(s_t)$ (<i>diagnostic</i>)	nearly flat in ϱ , what not to use here

A trap: $\text{Var}(\log z) = \sigma^2/(1 - \varrho^2)$ increases in ϱ , but $\text{Var}(\Delta \log z) = 2\sigma^2/(1 + \varrho)$ decreases.

And be clear what the “data” are: one simulation of the surrogate itself at $\varrho = 0.90$, under the *same* shock path as every candidate. Deliberate: it isolates identification from sampling noise, but the criterion then hits exactly zero. **This validates the machinery, not the estimator’s finite-sample behavior.**

Step 3: Minimize the Criterion

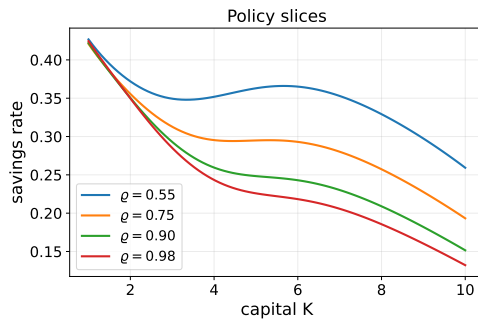
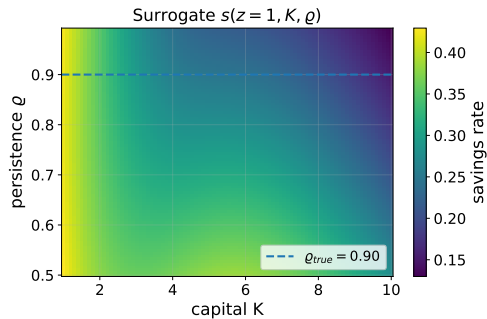
The scalar case, $\theta = \varrho$

$$\hat{\varrho} = \arg \min_{\varrho} [m(\varrho) - \hat{m}^{\text{data}}]^{\top} W [m(\varrho) - \hat{m}^{\text{data}}], \quad W = I.$$

1. One routine $\varrho \mapsto (C_t, I_t, Y_t) \mapsto m(\varrho)$, reusing the trained surrogate throughout.
2. Evaluate on a fine grid over $[0.50, 0.99]$, step 0.005, 99 points. **At surrogate speed this is cheaper than one call to the original model.**
3. Take the grid arg min. No separate optimizer is needed, and you see the whole criterion rather than one path through it.
4. **Check the solution is interior.** A corner solution invalidates the asymptotics we just set up, and usually signals a mis-specified moment.

$W = I$ is inefficient but consistent; the efficient $W^* = \hat{\Sigma}_g^{-1}$ changes the standard errors, not the argument.

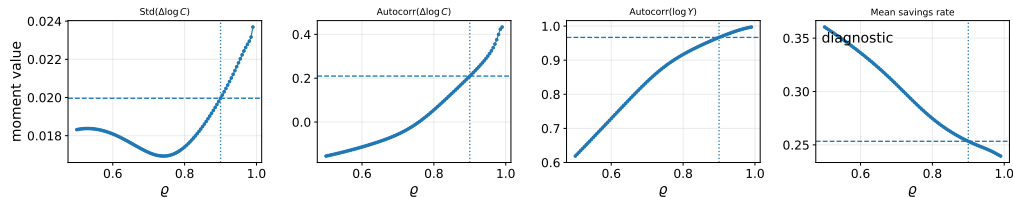
Result: The Surrogate Slice $s(K, \varrho \mid z = 1)$



One trained object standing in for 99 separate model solutions. Note what it does *not* show: the policy depends strongly on K and only **weakly** on ϱ : the four slices differ by only a few per cent at low K , and $\varrho = 0.90$ and 0.98 nearly coincide. That is not a defect. Most of the identifying information sits in the *transition*, not the policy: ϱ sets the persistence of z , and the moments are autocorrelations of simulated paths.

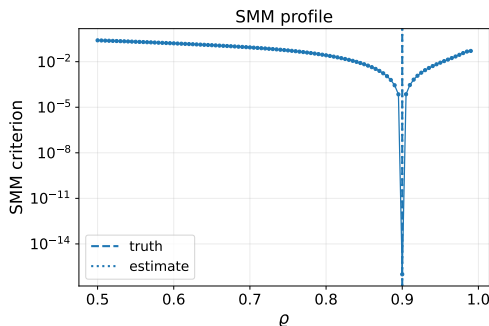
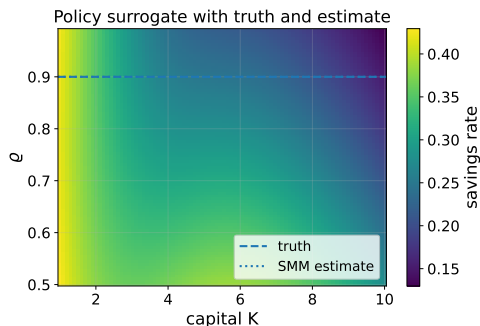
Result: Which Moments Actually Identify

Identifying variation under common random numbers



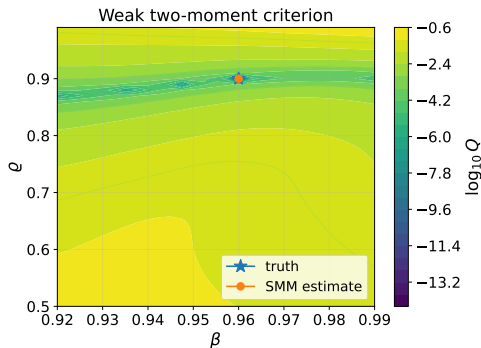
m_2 and m_3 rise monotonically in ρ : either alone would pin the parameter down. m_1 is **U-shaped**: nearly flat below $\rho \approx 0.75$, steep above it, so it identifies well near the truth and barely at all in the lower half of the range. The diagnostic m_0 is **essentially flat**: its measured slope at ρ_{true} is -0.07 , against $+1.02$ for m_2 . A moment with no slope contributes a near-zero column to M , and therefore almost nothing to identification, however well it matches in the data.

Result: The Criterion and the Estimate



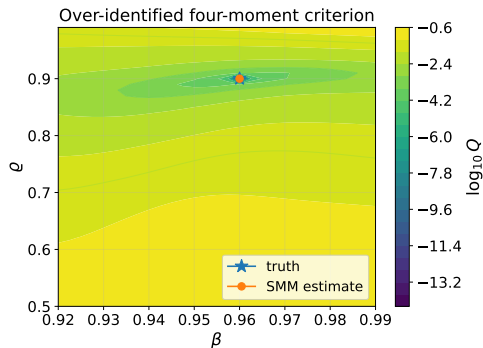
A single sharp interior minimum at $\hat{\rho} = 0.900$, exact because of the design on the previous slide, not because of luck. Note the *log* vertical axis: the criterion falls by three orders of magnitude between $\rho = 0.8$ and the minimum. **Curvature this steep is what good identification looks like**, but with a zero-noise design we cannot turn it into a standard error. Contrast the next slide.

Joint Estimation of (β, ϱ) : A Ridge, and How It Closes



Two dynamic moments. A long valley along β : many (β, ϱ) pairs fit about equally well. **Condition number 96**, smallest singular value 0.021.

The classical result: β moves the capital stock's *level*, so a level moment identifies it and business-cycle moments do not. **Identification is fixed by information, not by a better optimizer.**



Add the level moments. The valley closes into a basin. **Condition number 2.5**, smallest singular value 0.91.

The Ridge That Wasn't

An earlier run of this same notebook produced a ridge that **stayed open** when the level moments were added: condition number 305, the weak direction lying almost exactly along the β axis. It looked like a clean econometric finding. It was not.

The closed form gives $ds^*/d\beta = 1.92$. The under-trained surrogate reported -0.0016 : **near zero and the wrong sign**. Its Euler loss looked fine.

	200 steps	8,000 steps
s at the anchor	0.336	0.254
error vs. $s^* = 0.254$	32%	0.0%
$\partial m_0 / \partial \beta$	-0.0016	+1.27
condition number	305	2.5

The check that catches it

A flat direction is either the moments or **the surrogate**, and the criterion cannot tell you which. Evaluate the surrogate against something known in closed form first: here $s^*(\beta)$ at the steady state. This is *lazy learning* from this morning's Part VI, caught in the act.

From Classroom to Production

Brock–Mirman is the mechanism in miniature. Chen, Didisheim & Scheidegger (2026) run it on a high-dimensional workhorse option-pricing model.

What the surrogate enables

- ▶ Re-estimate **every day** instead of once per paper
- ▶ Systematic out-of-sample evaluation against reduced-form benchmarks
- ▶ Conditional distributions of option returns

What that produced

- ▶ An option-implied **tail-risk index**, predictive of market crashes
- ▶ A **parameter-instability** measure that lines up with option-market illiquidity
- ▶ A read on the structural-versus-reduced-form trade-off

The pattern worth remembering

None of these are faster versions of old results. They are questions whose *answers require a time series of estimates*, and that is exactly what a parametric surrogate manufactures.